



Intrinsic and radiation-induced defects in monoclinic ZrO_2 : Polaron mechanism of defect formation

Taras L. Petrenko^{a,*}, Igor P. Vorona^a, Valentyna V. Nosenko^{a,b}, Larysa Yu. Khomenkova^{a,b}

^a V. Lashkaryov Institute of Semiconductor Physics, National Academy of Sciences of Ukraine, 45 Pr. Nauky, Kyiv, 03028, Ukraine

^b National University "Kyiv-Mohyla Academy", Faculty of Natural Sciences, 2 Skovorody str., Kyiv, 04070, Ukraine

ARTICLE INFO

Keywords:

m-ZrO₂
DFT+U
Radiation defects
Polarons
Oxygen vacancies
Frenkel pairs

ABSTRACT

Mechanisms of radiation-induced defects formation in m-ZrO₂ powders were investigated using DFT+U calculations. In unirradiated crystal, oxygen vacancies exist in doubly charged state (V_O^{2+}) in threefold coordinated lattice sites together with two remote electron polarons. As a consequence, additional configurational entropy leads to considerable increase in vacancy and polaron concentrations. After crystal irradiation, the Frenkel pairs $V_O^{2+} - O_i^{2-}$ and electron-hole polaron pairs are formed. Interaction of such quasiparticles with Frenkel pairs leads to formation of the long-lived metastable centers observed in EPR experiment. Various charge states of oxygen vacancies were represented as complexes of doubly charged vacancy and one, two or three electron polarons. The most stable is complex V_O^+ with one trapped polaron, while concentration of another paramagnetic state of vacancy V_O^- is negligibly small at room temperature. Concentration of equilibrium electron polarons (Zr^{3+} centers) is sufficiently decreased after irradiation because such polarons are captured by nonequilibrium V_O^{2+} vacancies of Frenkel pairs. O^- centers were identified as the interstitial oxygen which traps the hole polaron. Electron-hole polaron pairs in irradiated m-ZrO₂ itself are short-lived defects and cannot be observed in experiment. However, interaction of polarons with native defects and impurities is important for formation of long-lived defects. In particular, we elucidate the nature of radiation-induced activation of phosphorus impurity. We show that in the ZrO₂ doped with phosphorus the self compensation occurs. After irradiation, polaron is captured by P_{Zr}^+ that converts this defect into paramagnetic state P_{Zr}^0 .

1. Introduction

Zirconium oxide has remarkable optical, mechanical, thermal, electrical and catalytic properties (Kouva et al., 2015; Chauhan et al., 2021; Hussien et al., 2025). Its excellent corrosion and radiation resistance allows it to be considered as a promising material for use in nuclear reactors (Gertsman et al., 1999; Motta et al., 2015). This application of ZrO₂ requires deep knowledge of the nature and mechanisms of radiation damage in it. The latter at the initial stage appear as point defects, which in paramagnetic state can be studied by electron paramagnetic resonance (EPR) technique. Further accumulation of the point defects, their interactions and evolution lead to significant deterioration of the mechanical properties of the material until its complete destruction. This determines the importance of studying radiation-induced defects.

The defects in ZrO₂ were observed using various techniques, in particular thermo and optical stimulated luminescence (Costantini et al., 2011), optical absorption (Khomenkova et al., 2024) and others. But the most detailed information about intrinsic and radiation-induced defects was obtained by EPR (Nosenko et al., 2026 and references

therein). It was found that the main intrinsic defects in ZrO₂, regardless of its crystal structure (cubic, tetragonal or monoclinic), are Zr^{3+} -based paramagnetic centers (isolated Zr^{3+} ions or its complex with one or two oxygen vacancies). The main radiation-induced paramagnetic centers (PCs) are singly charged oxygen vacancies (V_O^+) and hole trapped centers (O^-). However, a number of unresolved issues remain, in particular the localization of these defects in the crystal lattice, as well as their precursors and formation mechanisms. Theoretical calculations can provide essential assistance in this.

Intrinsic defects and polarons were subject of first-principle investigation during last two decades. In particular with the use of density functional theory (DFT) free of self interaction error it was shown that in m-ZrO₂ hole polarons are localized on the threefold O3 sublattice only (McKenna et al., 2012). The existence of electron polarons trapped on Zr^{4+} (in other terms Zr^{3+} PCs) in tetragonal crystal modification (t-ZrO₂) was predicted with B3LYP hybrid DFT calculations (Gionco et al., 2013).

Oxygen vacancy in metal oxides is common defect with widely varied concentration which is important for various applications (Lany,

* Corresponding author.

E-mail address: petrenkotl@isp.kiev.ua (T.L. Petrenko).

<https://doi.org/10.1016/j.radphyschem.2026.113954>

Received 19 February 2026; Received in revised form 17 April 2026; Accepted 20 April 2026

Available online 22 April 2026

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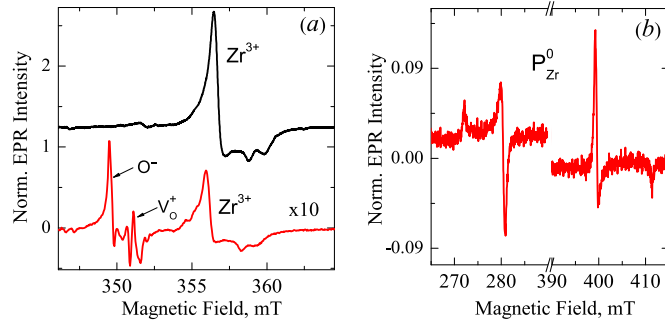


Fig. 1. EPR spectra of m-ZrO₂ powder before (black line) and after (red line) electron irradiation: (a) - intrinsic defects and (b) - phosphorus related spectra.

2015; Zhuang et al., 2020; Ullah et al., 2023; Gunkel et al., 2020; Hu et al., 2023; Seeman et al., 2020). In this paper we performed theoretical modeling of defects observed in unirradiated and electron-irradiated m-ZrO₂ powders (Nosenko et al., 2026). An important feature of such defects is that radiation-induced imperfections are not formed under thermodynamic equilibrium conditions, but are metastable ones. In this case we can only talk about local equilibrium conditions and standard methods for calculation of the intrinsic defects concentrations are not applicable. Moreover, important property of radiation defect is its stability over time, regardless of whether the total energy is higher or lower. Therefore, a situation is possible when a defect configuration with higher energy is experimentally observed.

It is well known that in a number of oxides electrons and holes exist as the localized polaron states rather than delocalized band carriers (Lany, 2015). However, at present interaction of radiation-induced polarons with both intrinsic and extrinsic defects is not sufficiently investigated. The present paper is devoted to modeling of metastable paramagnetic defects investigated in electron-irradiated m-ZrO₂ powders (Nosenko et al., 2026), its charge states and mechanisms of formation.

2. EPR spectra of initial and irradiated ZrO₂ powders

Experimental room temperature EPR spectra of initial and irradiated ZrO₂ powders are shown in Fig. 1. EPR spectra of initial ZrO₂ powders reveal signals originated from Zr³⁺ ions only. Irradiation of powders leads to appearance the new EPR signals resulted from V_O⁺, O⁻ and P_{Zr}⁰ paramagnetic defects (for details see Nosenko et al., 2026). At the same time a considerable decreasing of Zr³⁺ ions concentration is observed. In the present work we will consider intrinsic defects in unirradiated ZrO₂ as well as clarify the radiation defects models and formation mechanisms of such defects. For this purpose, a DFT study of defects involved in the described above EPR experiment was performed as described in the next section.

3. Method

To elucidate the nature of defect induced by irradiation of ZrO₂ crystal we performed DFT calculations using CP2K code (Kuehne et al., 2020). For description of small polarons hybrid functionals or DFT+U method are utilized as a rule. We used the latter one because in this case much less computing power is required. We employ PBE+U functional (Perdew et al., 1996) together with relativistic small core pseudopotentials (Goedecker et al., 1996; Hartwigsen et al., 1998; Krack, 2005). We choose parameter U = 7 eV for oxygen p-states and U = 10 eV for zirconium d-states. In this case calculated forbidden gap is 5.6 eV close to the experimental one. The Gaussian and plane waves method (GPW) (Lippert et al., 1997; VandeVondele et al., 2005) in conjunction with optimized short range Gaussian basis of double

zeta quality (VandeVondele and Hutter, 2007) was used. For Zr atom 4s²4p⁶5s²4d² electrons were considered as the valence ones and polarization f-functions were added to the basis. The multigrid with 750 Ha for density cutoff and 25 Ha for relative density cutoff were applied for calculations. Integration over Brillouin zone was performed with 2 × 2 × 2 Monkhorst-Pack grid (Monkhorst and Pack, 1976).

The Gaussian augmented plane wave method (GAPW) (Lippert et al., 1999) with all electron Ahlrichs def2-TZVP basis (Weigend and Ahlrichs, 2005) for phosphorus impurity was utilized for calculation of hyperfine interaction tensor. The 2 × 2 × 2 supercell (96 atoms) based on the conventional unit cell with $a = 10.308$ Å, $b = 10.456$ Å, $c = 10.659$ Å, and $\alpha = \gamma = 90^\circ$, $\beta = 99.437^\circ$ was used for calculations (Jain et al., 2013).

To estimate the contribution of artificial interaction between charged defects due to finite size of supercell, we use correction term (Makov and Payne, 1995) for calculated total energies:

$$\Delta E_{corr} = \frac{\alpha q^2}{2\epsilon(V_{SC})^{1/3}}. \quad (1)$$

Here $\alpha = 2.8373$ is Madelung constant for cubic lattice, q is charge of defect, ϵ is static dielectric constant of material and V_{SC} is a volume of the used supercell. For monoclinic ZrO₂ dielectric constant $\epsilon = 19.7$ (Zhao and Vanderbilt, 2002) is relatively large and $\Delta E_{corr} = 1.57$ eV for $q = 4$. This correction term is added to the values of total energies for charged defects in all cases. In Tables 2 and 3 (vide infra) we show both corrected and uncorrected values in order to be sure that such correction do not qualitatively changes results of calculations.

4. Polarons in monoclinic ZrO₂

In this section we consider polarons in monoclinic ZrO₂ produced by high energy irradiation. Crystal structure of m-ZrO₂ belong to the space group $P2_1/c$ with two inequivalent oxygen positions designated as O3 and O4. In position O3 oxygen is three-fold coordinated by Zr atoms, while O4 positions are four-fold coordinated lattice sites. Free hole polarons in m-ZrO₂ were investigated with DFT free of self interaction error (McKenna et al., 2012). The main features of such polarons important for investigation of radiation-induced defects are localization of holes mostly in O3 sublattice and low barrier (0.13 eV) for polaron diffusion. In this work rapid diffusion of hole polarons between O3 lattice sites in m-ZrO₂ was predicted.

Our calculations show that in m-ZrO₂, both electron polaron (e_p) and hole polaron (h_p) exist. Hole polarons, in general, may be localized both in O3 and O4 sites. The latter ones are higher in energy by 2.8 eV which qualitatively agree with conclusions of McKenna et al. (2012). For this reason, we will consider free hole polarons localized in the O3 lattice sites only. Electron polarons (e_p) are localized in the cation lattice sites (or Zr³⁺ PCs in other terms) and both notations are equivalent. In further analysis, we prefer to use the first notation (e_p).

After crystal irradiation, at first high energy band electrons and holes are formed. Next, thermalization hole and electron band states undergo transformation into polaron states because of decreasing the total energy as well as because of increasing the configurational entropy for polarons (Lany, 2015). In general, there may be a potential barrier between these states. To estimate the height of barrier, we performed linear interpolation of potential energy surface between supercell geometries for band and polaron states similar to McKenna et al. (2012):

$$\vec{R}(t) = t\vec{R}(h_p) + (1-t)\vec{R}_{band}^h \quad (2)$$

$$\vec{R}(t) = t\vec{R}(e_p) + (1-t)\vec{R}_{band}^e \quad (3)$$

Here $\vec{R}(h_e)$ and $\vec{R}(h_p)$ are positions of atoms in supercell with electron and hole polarons, $\vec{R}_{band}^e$ and $\vec{R}_{band}^h$ are atomic positions in supercells with band electrons and holes, respectively. Parameter t is

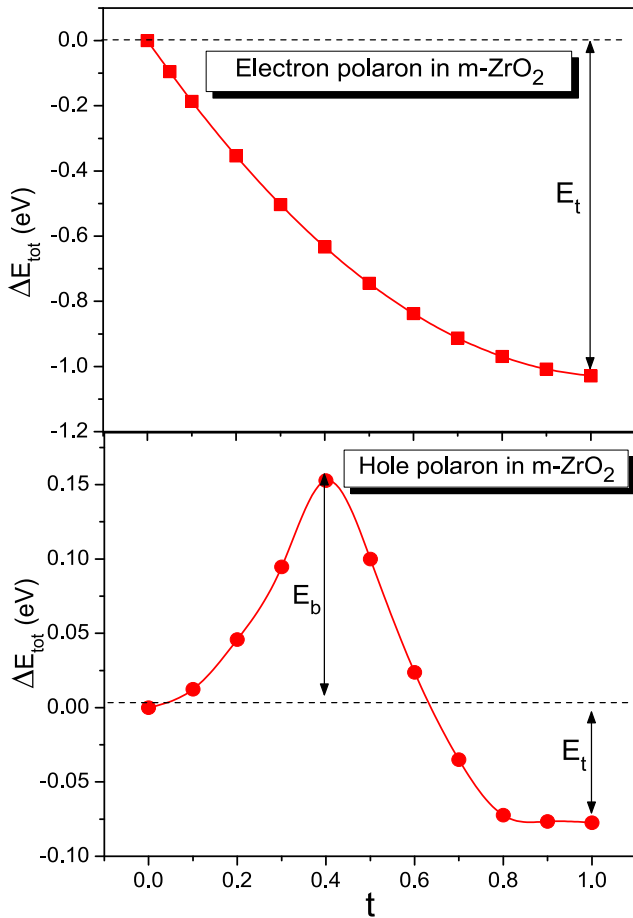


Fig. 2. Transition from band states ($t = 0$) to polaron ones ($t = 1$) for electrons and holes according to Eq. (2).

varied in the range of $0 \leq t \leq 1$, where $t = 0$ corresponds to the band geometry, while $t = 1$ - to the hole or electron polaron geometry. This simple approach gives the upper bound to the true potential barrier value and exact barrier height is somewhat lower.

Barrier height and hole polaron stabilization energy is in semiquantitative agreement with McKenna et al. (2012). Namely, we obtain barrier height $E_b = 0.15$ eV versus 0.05 eV and polaron stabilization energy $E_p = 0.08$ eV versus 0.13 eV in McKenna et al. (2012). For electron polarons transition from band states to localized polaron states is barrier free as it is follows from Fig. 2.

5. Polaron assisted oxygen vacancies formation before irradiation

Oxygen vacancy concentration in metal oxides depends on the number of factors. In particular, annealing at high temperatures and/or in vacuum leads to increasing of entropy for O_2 molecules in ambient atmosphere, followed by the increase of vacancy concentration. Besides, there is strong dependence of vacancy concentration on the concentration of purposeful or accidentally introduced acceptor impurities (Zhang et al., 2020): the more acceptor concentration the more vacancy concentration (Hagemann and Hennings, 1981). However, there is an important difference between these cases. In the first, one vacancy formation is accompanied with simultaneous appearance of electron polarons, which may be observed in EPR spectrum. In the second case, nonparamagnetic doubly charged oxygen vacancy is formed, while two electrons are localized on the acceptor level rather than exist as free polarons.

In this section, we will consider neutral oxygen vacancy taking into account electron configurational entropy contribution to the Gibbs free energy (Zhou et al., 2006; Lenarduzzi and Cooper, 2021) which dramatically decrease the effective formation energy of such vacancies.

5.1. Neutral oxygen vacancy as defect complex of charged vacancy and electron polarons

Oxygen vacancy with electron polarons located in cation sublattice is characterized by a significantly larger number of microstates compared with the vacancy alone. Namely, for certain vacancy position polarons may be located in a large number of different cation sites which increase the overall configurational entropy which in turn reduce the vacancy formation energy. To estimate quantitatively this effect in diluted limit (small vacancy and polaron concentrations) we used simplified model. Namely, all positions of polarons in cation sublattice are assumed to be equivalent and considered as free polarons despite the different distances between polaron and charged vacancies. Besides we ignored correlation between polaron positions when two negative polarons cannot be located close one to another in the crystal lattice.

In all cases we considered overall neutral systems, but parts of such systems are charged. Firstly, we consider neutral oxygen vacancy as complex of doubly charged (empty) vacancy itself and two polarons localized in remote cation sites of perfect lattice ($V_O^{2+} + 2e_p$). For calculation of formation energy of such system we used doubly charged supercell with oxygen vacancy and negatively charged supercell of initially perfect crystal in which electron polaron is formed after geometry optimization. Considerable lowering of formation energy in this case is caused by increasing of the number microstates due to additional configuration entropy associated with two polarons. Formation of such defect is accompanied by the variation of Gibbs free energy. In particular, if initial free energy for perfect lattice is $G_{ini} = 3n_v^{(2+)} \times E_{tot}^{perf}$ then after transferring of oxygen atom to the ambient atmosphere and formation of vacancy together with two remote polarons, final free energy G_{fin} may be calculated as

$$G_{fin} = n_v^{(2+)} [E_{tot}(V_O^{(2+)}) + \mu_O^{(g)}] + 2n_v^{(2+)} E_{tot}(e_p) - S_1 T - S_2 T, \quad (4)$$

where $n_v^{(2+)}$ is concentration of doubly charged oxygen vacancies, $E_{tot}(V_O^{(2+)})$ and $E_{tot}(e_p)$ are total energies of corresponding isolated vacancies and polarons, $\mu_O^{(g)} = \frac{1}{2} E_{tot}(O_2^{(g)})$ is chemical potential of oxygen in gaseous phase, and T is temperature. Vacancy-related and polaron-related configurational entropy terms S_1 and S_2 may be represented as

$$S_1 = k_B \ln[C_{N_{O_3}}^{n_v^{(2+)}}] = k_B \ln\left[\frac{N_{O_3}!}{n_v^{(2+)}!(N_{O_3} - n_v^{(2+)})!}\right] \quad (5)$$

$$S_2 = k_B \ln[C_{N_{Zr}}^{2n_v^{(2+)}}] = k_B \ln\left[\frac{N_{Zr}!}{(2n_v^{(2+)})!(N_{Zr} - 2n_v^{(2+)})!}\right] \quad (6)$$

Here N_{O_3} and N_{Zr} are densities of O_3 and cation sites in the crystal lattice, respectively, k_B is Boltzmann constant and concentration of polarons is $[e_p] = 2[n_v^{(2+)}]$. Equilibrium concentration of doubly charged oxygen vacancies $n_v^{(2+)}$ may be founded from the condition $\frac{\partial \Delta G}{\partial n_v^{(2+)}} = 0$, $\Delta G = G_{fin} - G_{ini}$. In such a way with the use of Stirlings approximation for $n!$ we obtain

$$n_v^{(2+)} = (N_{O_3} N_{Zr}^2)^{1/3} \times \exp[-E_{form}(V_O^{2+})/3k_B T] \quad (7)$$

$$E_{form}(V_O^{(2+)}) = E_{tot}(V_O^{(2+)}) + \mu_O^{(g)} + 2E_{tot}(e_p) - 3E_{tot}^{perf} \quad (8)$$

We note that concentration of vacancies in Eq. (7) is determined by the effective reduced formation energy $E_{form}(V_O^{2+})/3 = 2.46$ eV due to additional polaron-related entropy term S_2 .

Next we will consider the defect $V_O^+ + e_p$. In this case one polaron is localized on the oxygen vacancy (i.e on the nearest Zr atom) and the second polaron is localized on a distant cation sites (Zr^{3+}). For

calculation of formation energy of such system we used singly charged supercell with oxygen vacancy and negatively charged supercell of initially perfect crystal in which electron polaron is formed after geometry optimization. In this case formation energy is increased because contribution of polaron configuration entropy decreases compared with the case of $V_O^{2+} + 2e_p$ defect.

Formula for concentration of paramagnetic singly charged vacancies together with one remote electron polaron ($V_O^+ + e_p$) may be derived in the same way as Eqs. (7) and (8):

$$n_v^{(+)} = (N_{O3}N_{Zr})^{1/2} \times \exp[-E_{form}(V_O^+)/2k_B T] \quad (9)$$

$$E_{form}(V_O^+) = E_{tot}(V_O^+) + \mu_O^{(g)} + E_{tot}(e_p) - 2E_{tot}^{perf} \quad (10)$$

Here reduced effective formation energy is $E_{form}(V_O^+)/2 = 3.24$ eV owing to one-polaron contribution to configurational entropy.

We note that Eq. (10) is formally equivalent to well known formula for concentration of Frenkel pairs ($V_O + O_i$) in a crystal:

$$n(V_O + O_i) = (N_O N_i)^{1/2} \times \exp(-E_{form}/2k_B T) \quad (11)$$

Here, N_O and N_i are numbers of oxygen and interstitial sites in the volume unit. In this case reduction of formation energy is caused by additional configurational entropy associated with interstitial oxygen similar to the contribution of polarons.

For neutral vacancy when two electrons are localized on the nearest neighbor zirconium atoms and polaron-related entropy contribution vanishes, we obtain standard formula for vacancy concentration:

$$n_v^{(0)} = N_{O3} \times \exp[-E_{form}(V_O^0)/k_B T] \quad (12)$$

$$E_{form}(V_O^0) = E_{tot}(V_O^0) + \mu_O^{(g)} - E_{tot}^{perf} \quad (13)$$

In this case $E_{form}(V_O^0) = 6.06$ eV is too large and neutral vacancy concentration is negligible. We note that this value agree well with the value of 6.35 eV reported in Liu et al. (2014) for cubic ZrO_2 . Therefore, vacancy in m- ZrO_2 exists mainly as V_O^{2+} together with remote electron polarons localized on the zirconium lattice sites.

6. Intrinsic defects formation after irradiation

6.1. Interstitial oxygen

In general, after ZrO_2 crystal irradiation one may expect formation of metastable primary defects in the form of vacancy-interstitial pair V_O^{2+} and O_i^{2-} . Therefore, it is important to find position of interstitial oxygen in the crystal lattice after Frenkel pair formation. We considered several locations of interstitial oxygen including split-interstitial positions in vicinity of O3 and O4 sites.

After the hole polaron capture interstitial doubly charged oxygen O_i^{2-} transforms into singly charged O_i^- paramagnetic defect. Therefore, calculations were performed for charged supercells with $q = -2$ and $q = -1$ containing interstitial oxygen. We founded interstitial positions shown in Figs. 3 and 4 for O_i^- with lowest total energy and similar position was founded for O_i^{2-} . Split-interstitial configurations in O3 lattice sites were founded to be unstable and transforms into positions shown in Figs. 3 and 4 for O_i^- and O_i^{2-} . In O4 lattice positions stable split-interstitial configurations were founded for both charge states of interstitial oxygen. However, total energies of such local minima are higher by 3.6 eV (O_i^-) and 7.8 eV (O_i^{2-}).

In such a way we found interstitial position corresponding to the total energy minimum shown in Fig. 3. In this case local atomic arrangement resembles the one for oxygen atom in the regular lattice. In particular, interstitial oxygen is three-fold coordinated as well. Moreover Zr-O interatomic distances d_1 , d_2 and d_3 are close one to another as shown in Table 1. Therefore, interstitial oxygen after hole capture forms ionic bonds with the nearest Zr atoms. In this case hole diffusion is associated with hopping of oxygen atom between equivalent positions in crystal lattice accompanied with the bond breaking and with formation of relatively large barrier for migration. In contrast

Table 1

Values of Zr-O interatomic distances (in Å) for three-fold coordinated oxygens in Fig. 3.

Defect	d_1	d_2	d_3
O_i^-	2.17	2.24	2.32
h_p	2.17	2.28	2.21
Perfect lattice	2.16	2.06	2.07

migration of free hole polaron located in O3 sites occurs without breaking of chemical bonds leading to formation of lower barriers.

Potential energy barrier overcoming by interstitial oxygen (see Fig. 4) was calculated according to linear interpolation formula similar to Eq. (2):

$$\vec{R}(t) = t\vec{R}_1(O_i) + (1-t)\vec{R}_2(O_i) \quad (14)$$

where $\vec{R}_1(O_i)$ and $\vec{R}_2(O_i)$ are atomic positions of two equivalent minima of interstitial oxygen O_i^- and results of calculations are shown in Fig. 5. One can see that potential barrier for migration of this center is more then three times larger compared to the barrier height of 0.13 eV for hole polaron (McKenna et al., 2012).

Based on these arguments, we conclude that O_i^- is hole-related long-lived metastable radiation-induced defect which probably is observed in EPR experiment (Nosenko et al., 2026).

6.2. Interaction of polarons with intrinsic defects in m- ZrO_2

Another expected effect of irradiation is creation of electron-hole pairs which exist as electron and hole polarons. If such polarons are associated with a large lattice relaxation and are strongly localized, then existence of long-lived metastable defects may be expected. Otherwise, rapid electron-hole recombination prevents the formation such type of radiation defects. As mention above the barriers height for hole diffusion is low (does not exceed 0.13 eV). Therefore, rapid diffusion of hole polarons leads to a high probability of electron-hole pair recombination after irradiation. In this case long-lived metastable defects may be formed after interaction of free polarons with intrinsic defects, namely with oxygen vacancies V_O and oxygen interstitials O_i . Results of such calculations are presented in Table 2.

We pay attention to the defect reactions (items 7 and 8) indicated in Table 2. Formation energy of Frenkel pair $V_O^{2+} + O_i^{2-}$ is large enough (6.50 eV), however hole polaron assisted formation energy of $V_O^{2+} + O_i^-$ pair is sufficiently lower (4.38 eV). This means that energy cost for transition of oxygen atom from O3 lattice site, where hole polaron is localized, into the interstitial lattice position, is less then similar transition from the regular O3 lattice site. The reason is that in this case Zr-O bonds are electron deficient and weakened. Therefore, moving of oxygen atom from a lattice site to an interstitial position requires less energy. Substantial formation energy reduction is caused only by existence of hole polarons. Such polarons may be produced by various types of irradiations such as gamma, electron, ultraviolet etc. Probably this is the main way of formation the O_i^- paramagnetic radiation defect observed in our experimental part (Nosenko et al., 2026).

Another way of O_i^- defect formation is described by the two step reaction. The first step is conventional reaction which is described by item 7 in Table 2 resulting in formation of nonparamagnetic defect O_i^{2-} . The second one is indicated as item 9 in this table and O_i^- center is formed after capture of hole polaron by interstitial oxygen O_i^{2-} .

6.3. Oxygen vacancies as complexes of V_O^{2+} and electron polarons

There are three possible sources of doubly charged oxygen vacancies V_O^{2+} in m- ZrO_2 . The first one is considered above when neutral oxygen vacancy V_O^0 exists in the form of V_O^{2+} and two distant electron polarons at finite temperatures. The second one arises when there are relatively

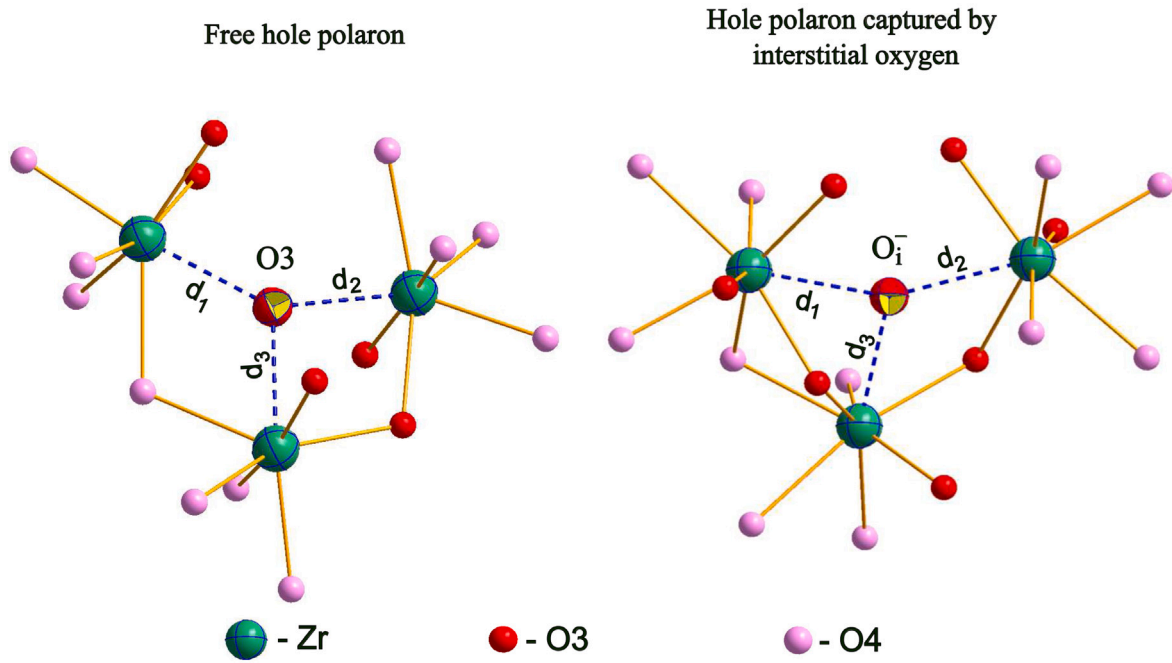


Fig. 3. Hole polaron (h_p) localized in O3 lattice position and O_i^- paramagnetic center. The latter one is interstitial oxygen O_i^{2-} which capture the hole polaron.

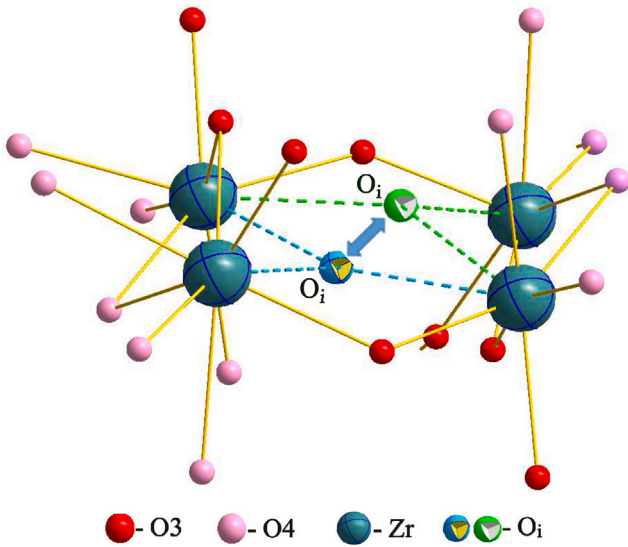


Fig. 4. Migration of interstitial oxygen O_i^- between equivalent interstitial positions in m-ZrO₂. Distance between such positions is equal to 1.36 Å.

shallow acceptor defects. These two sources work in conditions of thermodynamic equilibrium. The third source is formation of Frenkel pairs in nonequilibrium conditions after irradiation. Vacancies produced in the second and third ways may interact with nonequilibrium electron polarons that arise after irradiation. In this case various charge states of vacancies formally can be considered as vacancy-polaron complexes according to the following reactions:

$$V_O^{2+} + e_p \rightarrow V_O^+, \quad E_b^{(1)} = -0.89 \text{ eV} \quad (15)$$

$$V_O^{2+} + 2e_p \rightarrow V_O^0, \quad E_b^{(2)} = -1.31 \text{ eV} \quad (16)$$

$$V_O^{2+} + 3e_p \rightarrow V_O^-, \quad E_b^{(3)} = -1.96 \text{ eV} \quad (17)$$

Here $E_b^{(i)}$ ($i = 1 \div 3$) is binding energy of complexes and negative sign testify about stability of complexes. The question is which of these

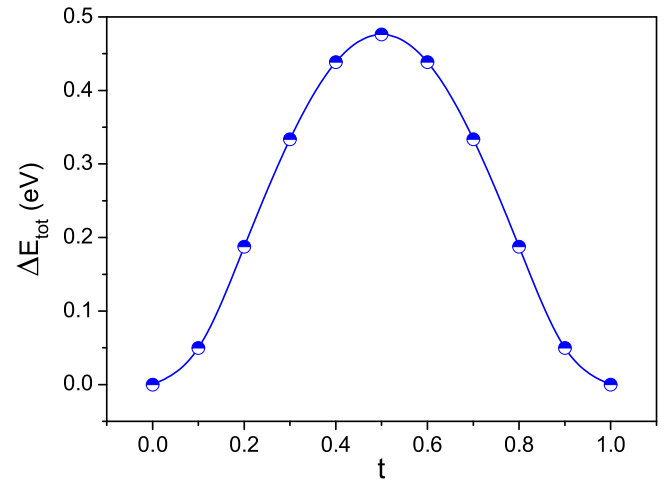


Fig. 5. Potential energy barrier for migration of interstitial oxygen between equivalent positions in m-ZrO₂ lattice as shown in Fig. 4.

complexes prevails at the actual temperatures. Paramagnetic states V_O^+ and V_O^- are of special interest because vacancy related centers in the electron irradiated ZrO₂ are observed in EPR experiment (Nosenko et al., 2026). It is important to note that in the case of V_O^+ the ground spin state is doublet ($S = 1/2$), while for V_O^- the spin quartet ($S = 3/2$) is the ground one. In the latter case the nearly all spin density is localized on the three nearest to vacancy Zr atoms. Our calculations show that quartet spin state for V_O^- is by 0.63 eV lower in energy compared to the doublet state.

Let us consider system of n_v^{tot} vacancies which includes various possible vacancy charge states (+2, +1, 0, -1) and electron polarons with total number of n_{pol}^{tot} . We suppose that there is quasiequilibrium and part of polarons may be captured by doubly charged vacancies V_O^{2+} . Namely, there are $n_1(V_O^+)$ vacancies captured one polaron, $n_2(V_O^0)$ vacancies captured two polarons and $n_3(V_O^-)$ vacancies captured three polarons. We also suppose that total concentrations of vacancies and polarons are

Table 2

Metastable defects formed in monoclinic ZrO₂ in nonequilibrium conditions (after irradiation). $\Delta E = E_{tot}^{fin} - E_{tot}^{ini}$ is the total energy gain as the result of respective solid state reaction and negative value of ΔE corresponds to the exothermic one. This value can also be considered as the binding energy of the respective complex. Designation *Perf* means perfect ZrO₂ crystal. All calculations were performed at the PBE+U level of theory with correction according to Eq. (1).

Reaction number	Solid state reaction Initial state → Final state	ΔE (eV)		Conclusions
		uncorrected	corrected	
1	$V_O^{2+} + e_p \rightarrow V_O^+ + Perf$	-0.50	-0.89	Nonequilibrium electron polaron formed after irradiation is captured by V_O^{2+} vacancy located in O3 sites. Paramagnetic V_O^+ center is observed by EPR experiment.
2	$V_O^{2+} + 2 \times e_p \rightarrow V_O^0 + 2 \times Perf$	-0.72	-1.31	Two nonequilibrium electron polaron are captured by V_O^{2+} vacancy located in O3 sites and nonparamagnetic V_O^0 center is formed.
3	$V_O^{2+} + 3 \times e_p \rightarrow V_O^- + 3 \times Perf$	-1.37	-1.96	Negative vacancy V_O^- may be considered as complex of V_O^{2+} and three electron polarons. Relatively small value of binding energy testify in favor of small concentration of V_O^- vacancy at room temperature compared to concentration of V_O^{2+} and V_O^+ centers.
4	$V_O^+ + e_p \rightarrow Perf + V_O^0$	-0.22	-0.42	This reaction results in formation of nonparamagnetic center. However, binding energy is too small and complex V_O^0 is mainly in the dissociated state. In this case capture of additional electron polaron by V_O^+ is improbable.
5	$V_O^+ + 2 \times e_p \rightarrow V_O^- + 2 \times Perf$	-0.86	-1.06	Negative vacancy V_O^- may be considered as complex of V_O^+ and two electron polarons. Relatively small value of binding energy testify in favor of small concentration of V_O^- vacancy compared to concentration of V_O^+ (see Section 6.3).
6	$V_O^0 + h_p \rightarrow Perf + V_O^+$	-3.26	-3.26	In this case binding energy is large and dissociation of paramagnetic V_O^+ complex into nonparamagnetic neutral vacancy and hole polaron is improbable.
7	$2 \times Perf \rightarrow V_O^{2+} + O_i^{2-}$	5.72	6.50	This is formation energy of Frenkel pair in the initially perfect lattice as a result of irradiation.
8	$Perf + h_p \rightarrow V_O^{2+} + O_i^-$	3.99	4.38	This is reduced formation energy of polaron-assisted creation of Frenkel pair $V_O^{2+} - O_i^-$ with paramagnetic interstitial oxygen
9	$h_p + O_i^{2-} \rightarrow Perf + O_i^-$	-1.73	-2.12	Free hole polaron is captured by interstitial oxygen of primary radiation defect ($V_O^{2+} - O_i^{2-}$). O_i^- is long-lived paramagnetic center which can be observed by EPR experiment.
10	$e_p + O_i^- \rightarrow Perf + O_i^{2-}$	-1.75	-1.56	Free electron polaron is captured by the paramagnetic interstitial oxygen O_i^- . The probability of this process is less than probability of hole polaron capture by O_i^{2-} because of lower binding energy (-1.56 eV vs -2.12 eV).

constants:

$$n_v^{tot} = n_0(V_O^{2+}) + n_1(V_O^+) + n_2(V_O^0) + n_3(V_O^-) \quad (18)$$

$$n_{pol}^{tot} = n_{pol}(free) + n_1(V_O^+) + 2n_2(V_O^0) + 3n_3(V_O^-) \quad (19)$$

Here $n_{pol}(free)$ is concentration of free polarons and for simplicity we put $n_v^{tot} = n_{pol}^{tot} = n_{tot}$. Under these assumptions concentrations of oxygen vacancies in various charge states may be determined from the following system of three equations of the mass-action law type (Van de Walle and Neugebauer, 2004; Petrenko et al., 2024) to find n_1 , n_2 and n_3 :

$$\begin{aligned} & [n^{tot} - n_1(V_O^+) - n_2(V_O^0) - n_3(V_O^-)] \\ & \times [n^{tot} - n_1(V_O^+) - 2n_2(V_O^0) - 3n_3(V_O^-)]^k = \\ & n_i(N_{Zr})^k \times \exp(E_b^{(i)}/k_B T), \quad i, k = 1, 2, 3, i = k \end{aligned} \quad (20)$$

where i is label and k is power. Binding energies $E_b^{(i)}$ are determined by Eqs. (15)–(17) and N_{Zr} is site density of Zr in m-ZrO₂ crystal. To estimate the degree of free polaron localization on the V_O^{2+} resulting in formation of paramagnetic defect V_O^+ we put $n_2 = n_3 = 0$ in Eq. (20). Fig. 6 shows calculated value of $n_1(V_O^+)/n_{tot}$ versus temperature. One can see that at room temperatures in wide range of vacancy concentrations we obtain $n_1(V_O^+) = n_{tot}$, i.e. all free electron polarons are captured by V_O^{2+} .

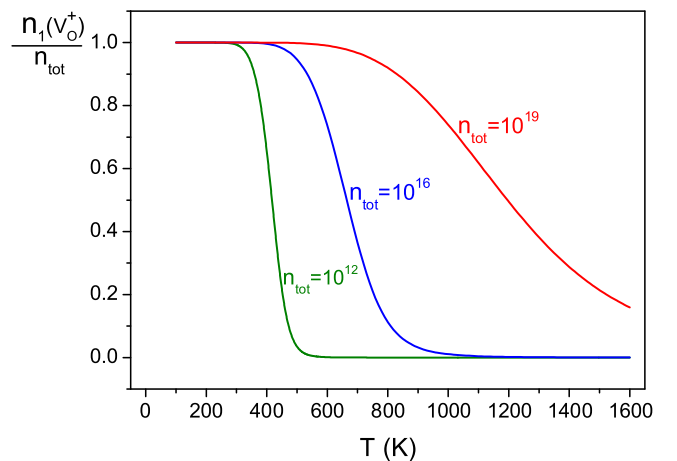


Fig. 6. Fraction of singly charged vacancies V_O^+ in quasiequilibrium conditions after irradiation versus temperature for various total amount of vacancies $n_{tot} = n_1(V_O^+) + n_0(V_O^{2+})$.

To estimate concentration of another possible paramagnetic state of vacancy V_O^- from Eqs. (20), we note that

$$\frac{n_3(V_O^-)}{n_1(V_O^+)} \leq \frac{n_{tot}^2}{N_{Zr}^2} \times \exp(E_b^{(1)} - E_b^{(3)})/k_B T = X_{up}(T) \quad (21)$$

In Eq. (21) X_{up} is upper bound to the true value of relative concentration $n_3(V_O^-)/n_1(V_O^+)$ of these defects. Our calculations give $X_{up} \ll 1$ at room temperature. Therefore, concentration of negatively charged vacancies at room temperature is small compared to positively charged ones. Additional argument in favor of V_O^+ over V_O^- identification in EPR spectrum of ZrO_2 is that in the latter case for spin $S = 3/2$ there must be additional splitting (the so called zero-field splitting) in the spectrum which is not observed in experiment. The existence of V_O^- vacancy state may be expected only in the thermodynamic equilibrium conditions when Fermi level is located close to the conduction band minimum.

Thus, we identify vacancy related center observed by EPR, as singly charged oxygen vacancy V_O^+ . At the same time at high temperatures we obtain negligible concentration of this defect due to large contribution of electron entropy terms (see Eq. (5)).

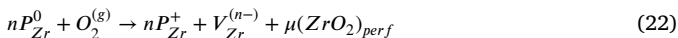
6.4. Density of states induced by intrinsic defects in m-ZrO₂

It is obvious that there are two ways for degradation of long-lived metastable radiation defects. The first one is caused by diffusion of radiation defects over crystal. In our case migration of interstitial paramagnetic oxygen O_i^- results in recombination with paramagnetic oxygen vacancy. Another way of EPR signal degradation is possible through solid state reaction shown in Table 2 (item 1). In this case electron from the V_O^+ defect must be emitted into the conduction band with subsequent transformation into polaron which in turn is captured by interstitial oxygen O_i^- and form nonparamagnetic defect O_i^{2-} .

In Fig. 7, calculated total density of states (DOS) induced in otherwise perfect crystal by actual paramagnetic defects is shown. One can see that in all cases radiation defects introduce a very deep states inside band. Therefore, thermal emission of electrons into the conduction band is improbable which suggests the stability of such defects.

7. Nature of radiation-induced activation of phosphorus impurity in m-ZrO₂

It is well known (Look et al., 2011; Mao et al., 2018; Swallow et al., 2018) that in many oxides doping with impurities may lead to formation of native defects and in a such a way to the self-compensation effect. To explain the absence of EPR spectrum in phosphorus doped ZrO_2 crystal before irradiation, we consider reaction in the defect subsystem and corresponding formation energy E_{form} for quadruple charged ($n = 4$) and doubly charged ($n = 2$) cation vacancies V_{Zr}^{n-} . In the case of O-rich growth conditions we have



$$E_{form}(V_{Zr}^{n-}) = n \times E_{tot}(P_{Zr}^+) + E_{tot}(V_{Zr}^{n-}) + \mu(ZrO_2)_{perf} - [n \times E_{tot}(P_{Zr}^0) + E_{tot}^{perf} + \mu(O_2^{(g)})] \quad (23)$$

Here external $O_2^{(g)}$ denotes oxygen molecules in gaseous form and μ is chemical potential of molecule in gaseous reservoir, E_{tot}^{perf} is total energy of perfect crystal lattice. The physical meaning of the last term in Eq. (22) is as follows. The Zr atom leaves the cation lattice site and left behind the vacancy, then Zr binds with O_2 molecule from reservoir. In a final step, such ZrO_2 fragment is embedded in the perfect crystal lattice as the building block. Chemical potential of this fragment was calculated as $\mu(ZrO_2)_{perf} = E_{tot}^{perf}/32$ in accordance with the used 96 atoms supercell. Obtained results are shown in Table 3.

One can see that in oxygen-rich environment, formation of quadruple charged zirconium vacancy is preferable compared to doubly charged

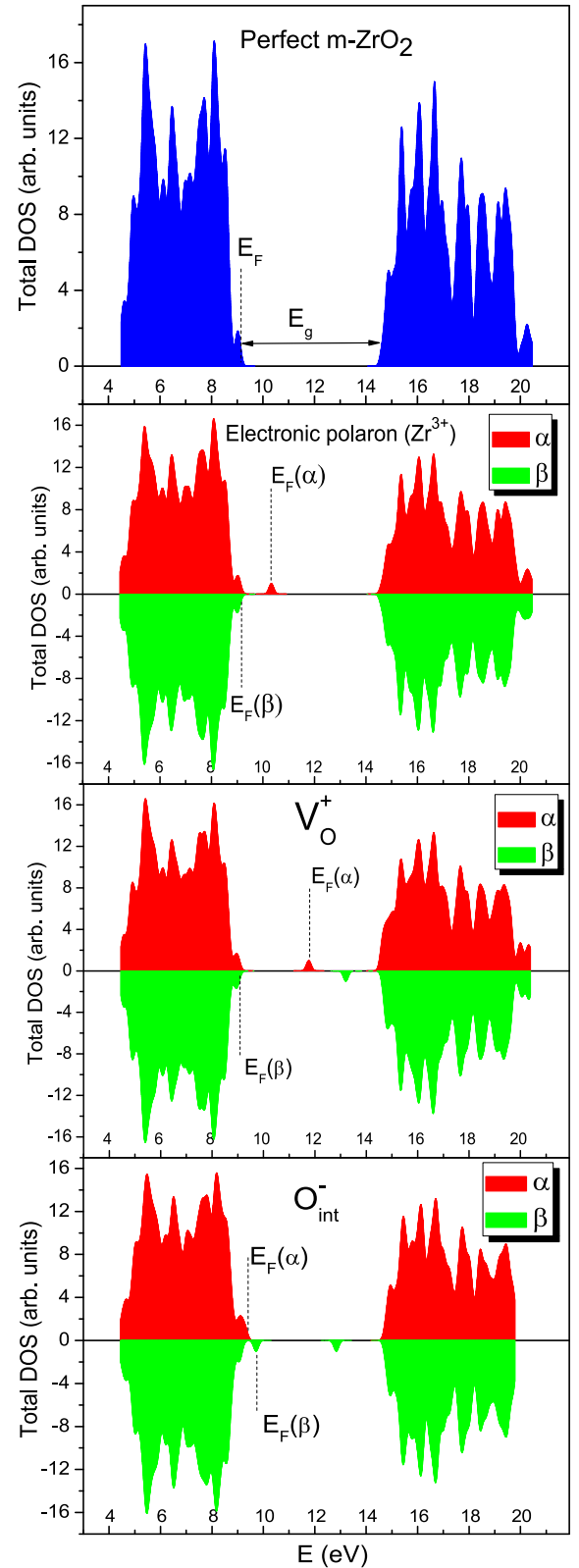


Fig. 7. Additional total density of states (DOS) induced by paramagnetic intrinsic defects in the band gap compared with perfect m-ZrO₂. DOS for α and β electrons (spins $+1/2$ and $-1/2$) are marked with red and green, respectively. For pictorial illustration we conditionally attribute a negative density of states to beta electrons.

Table 3

Defects formed in phosphorus doped monoclinic ZrO_2 before and after irradiation according to DFT calculations. $\Delta E = E_{\text{tot}}^{\text{fin}} - E_{\text{tot}}^{\text{ini}}$ is the total energy gain as the result of respective solid state reaction and negative signs of ΔE correspond to the exothermic reactions. Values of ΔE corrected for elimination the artificial defect–defect interaction in supercell according to Eq. (1) are shown together with uncorrected quantities.

Reference number	Solid state reaction Initial state $\rightarrow$ Final state	Method	ΔE (eV)	
			uncorrected	corrected
1	$4\text{P}_{\text{Zr}}^0 + \text{Perf} + \text{O}_2^{(g)} \rightarrow 4\text{P}_{\text{Zr}}^+ + \text{V}_{\text{Zr}}^{4-} + \mu(\text{ZrO}_2)$	PBE PBE+U	−4.74 −3.55	−2.77 −1.58
2	$2\text{P}_{\text{Zr}}^0 + \text{Perf} + \text{O}_2^{(g)} \rightarrow 2\text{P}_{\text{Zr}}^+ + \text{V}_{\text{Zr}}^{2-} + \mu(\text{ZrO}_2)$	PBE PBE+U	0.81 2.32	1.40 2.91
3	$e_p + \text{P}_{\text{Zr}}^+ \rightarrow \text{Perf} + \text{P}_{\text{Zr}}^0$	PBE+U	−0.76	−0.96

Table 4

Calculated main values of hyperfine interaction tensors (in 10^{-4} cm^{-1}) for neutral phosphorus impurity P_{Zr}^0 substituted for zirconium and located in two crystallographically inequivalent oxygen sites - O3 and O4.

Position in Lattice	Functional	A_{xx}	A_{yy}	A_{zz}
P_{Zr}	PBE	1022	864	863
P_{Zr}	PBE+U	1050	872	870
P_{O3}	PBE	207	−17	−18
P_{O4}	PBE	192	−17	−18
Exp. Nosenko et al. (2026)		1265	1075	1075

one. Moreover, concentration of $\text{V}_{\text{Zr}}^{4-}$ is exactly equal to $[\text{P}_{\text{Zr}}]/4$ because corresponding reaction in Table 3 is exothermic one and cation vacancies are formed spontaneously.

Mechanism of radiation response of ZrO_2 doped with phosphorus is as follows. Before irradiation all phosphorus impurities are compensated by zirconium vacancies and EPR signal is not observed. After irradiation, the electron–hole polaron pair is formed and electron polaron is captured by singly charged phosphorus P_{Zr}^+ ($\text{P}_{\text{Zr}}^{5+}$ oxidation state) which leads to formation of metastable paramagnetic center P_{Zr}^0 (oxidation state $\text{P}_{\text{Zr}}^{4+}$) observed in experiment (Nosenko et al., 2026). Results of calculations shown in Table 3 (item 3) suggest that this process is energy profitable.

To substantiate this model, we performed calculation of hyperfine interaction tensors with phosphorus nucleus for various lattice positions and compared theoretical values with experiment (Table 4). There is a good agreement of calculated hyperfine parameters for neutral phosphorus substituted for Zr site with the experimental ones. At the same time, for hypothetical substitution model P_{O}^0 the pronounced difference between calculated and observed hyperfine parameters was found as shown in Table 4.

Total density of states in the case of phosphorus impurity doping is shown in Fig. 8. Here P_{Zr}^+ is nonparamagnetic impurity state before irradiation. After irradiation and polaron capture, this defect is recharging to the P_{Zr}^0 paramagnetic state, and deep donor state appears near the top of valence band as illustrated in Fig. 8.

Similar effects may be expected for isovalance donor impurities such as As and Sb in m-ZrO_2 . In this case, a self-compensation most probably occurs due to spontaneous formation of zirconium vacancy, as well as paramagnetic state can be activated by irradiation.

8. Conclusions

DFT+U calculations suggest that under irradiation both electron and hole polarons are formed and these polarons play important role in the formation of radiation defects. Nevertheless, such radiation-induced polarons themselves do not produce long-lived metastable defect states. However, such type of defects may appear after the capturing of polarons by intrinsic defects as well as by impurities.

In unirradiated ZrO_2 at the thermodynamic equilibrium, formation of oxygen vacancies is accompanied with two equilibrium electron polarons (Zr^{3+} centers). In this case, concentration of various charge states

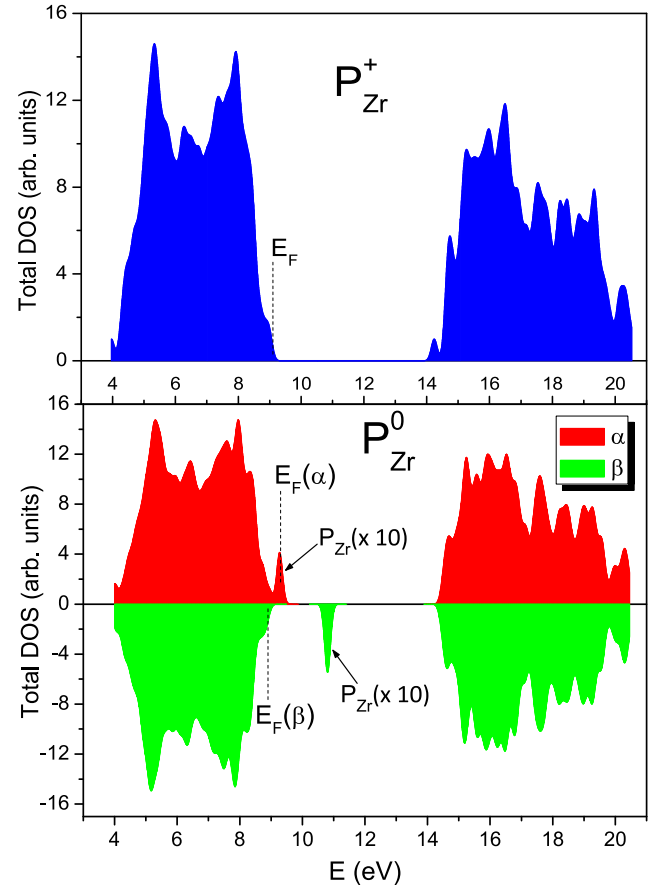


Fig. 8. Total density of states (DOS) for P_{Zr} impurity in m-ZrO_2 . P_{Zr}^+ is non-paramagnetic state caused by existence of compensating zirconium vacancy, while paramagnetic metastable P_{Zr}^0 impurity appear after electron polaron capture under irradiation. Projected DOS for phosphorus atom were multiplied by the factor of 10 for the sake of clarity. Total DOS for α and β electrons (spins $+1/2$ and $-1/2$) are marked with red and green, respectively. For pictorial illustration we conditionally attribute a negative density of states to beta electrons.

of oxygen vacancy is determined, among other things, by additional polaron related electron configurational entropy. This decreases considerably the vacancy formation energy in thermodynamic equilibrium. Our calculations show that in this case in m-ZrO_2 oxygen vacancies prefer to be empty (the V_{O}^{2+} charge states) and to be located in the threefold coordinated O3 lattice sites.

Other sources of doubly charged oxygen vacancies are Frenkel pairs ($\text{V}_{\text{O}}^{2+} - \text{O}_i^{2-}$) produced by electron irradiation of ZrO_2 . After the electron polaron capturing, an oxygen vacancy is recharged to the V_{O}^+ paramagnetic state revealed in EPR study. It is important that

doubly charged vacancies of Frenkel pairs do not accompanied with electron polarons in contrast to oxygen vacancies formed in equilibrium conditions. Therefore, such nonequilibrium vacancies may trap nonequilibrium electron polarons and form metastable defects. At the same time, interstitial oxygen O_i^{2-} transforms into paramagnetic center O_i^- after the hole polaron capture. Calculations show that formation of $(V_O^{2+} - O_i^-)$ pair may be polaron assisted. Consequently, a significant decrease in the formation energy of Frenkel pair and lowering of the threshold of displacement energy for oxygen atom occur.

Formation of Frenkel pairs in irradiated m-ZrO₂ is in line with experimental fact (Nosenko et al., 2026): concentration of Zr³⁺ paramagnetic centers is significantly reduced after irradiation. The reason is in the capture of these polarons by V_O^{2+} vacancies of Frenkel pairs.

Calculations suggest that paramagnetic negative vacancy V_O^- may be formed as metastable complex $V_O^{2+} + 3e_p$. However, binding energy of such complex is not very large. In this case, entropy contribution dominates and leading to much lower concentration of V_O^- vacancies compared to the V_O^+ ones.

Microscopic model of phosphorus impurity P_{Zr} observed in m-ZrO₂ powders was elucidated as well. We revealed that presence of this impurity leads to spontaneous formation of cation vacancies V_{Zr}^{4+} which is the self-compensation phenomenon. In this case, phosphorus is in the nonparamagnetic state P_{Zr}^+ . After irradiation, this impurity captures the electron polaron and becomes the paramagnetic deep defect P_{Zr}^0 .

Finally, we can conclude that to provide reliable description of radiation-induced defect formation in m-ZrO₂, the existence of hole and electron polarons should be taken onto account. For other related compounds such as metal oxides a similar approach may be appropriate. However, various materials may differ in the polaron properties and in values of binding energies of polarons to vacancies. For instance, if polarons are associated with large local atomic relaxation and large barriers for diffusion, than such polarons itself may form long-lived metastable defects. At the other hand, if binding energies are sufficiently large than charged vacancy may capture three polarons leading to formation of e.g. the V_O^- paramagnetic centers instead of V_O^+ . In such a way, proposed approach can be employed for description of radiation defect formation in the variety of oxide compounds.

CRediT authorship contribution statement

Taras L. Petrenko: Writing – original draft, Resources, Investigation, Conceptualization. **Igor P. Vorona:** Writing – review & editing, Software, Investigation, Conceptualization. **Valentyna V. Nosenko:** Writing – review & editing, Visualization, Validation, Methodology, Investigation. **Larysa Yu. Khomenkova:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study is supported by Presidium of NAS of Ukraine in the framework of the implementation of the projects: "New physical principles and technologies for the development of the element base of modern photoelectronics in the infrared range", № 0125U000799 and "Fundamental physical processes underlying photoinduced control of the functional properties of semiconductors and dielectrics", Reg. No. III-1-26.

Data availability

No data was used for the research described in the article.

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